

TalentAI Engineering Constitution

This document will define every rule of our project.

Chapter 1

Mission

Why are we building TalentAI?

Not for college.

Not just for GitHub.

We are building TalentAI as if we are a startup creating an enterprise SaaS platform that solves real hiring problems using AI.

Chapter 2

Core Principles

These principles can never be broken.

Principle 1

Understand before coding.

Never copy code without understanding it.

Principle 2

Blueprint first.

Code second.

Principle 3

Everything is modular.

Nothing is tightly coupled.

Principle 4

Security is never optional.

Principle 5

Every feature must solve a real user problem.

Principle 6

AI assists users.

AI never makes irreversible decisions automatically.

For example:

AI may recommend candidates.

It will never automatically reject them.

A recruiter makes the final decision.

Principle 7

Every feature must be documented.

Principle 8

Code should be readable.

Not clever.

Principle 9

Performance matters.

Principle 10

The system should be scalable.

Chapter 3

Technology Principles

Frontend

React

Next.js

TypeScript

Tailwind

Backend

Java

Spring Boot

Database

PostgreSQL

Redis

Qdrant

AI

Spring AI

Ollama

Cloud

Free tier

Everything selected because it solves a problem—not because it's trendy.

Chapter 4

AI Principles

AI should always explain its output where practical.

Example:

Instead of

Resume Score

80

AI explains:

Missing Docker

Weak project descriptions

Add measurable achievements

Improve summary

The user understands why.

Chapter 5

Security Principles

Passwords

Never stored in plain text.

JWT

Always validated.

Files

Always validated.

Role checks

Always enforced.

Input

Always validated.

Logs

Never contain passwords or secrets.

Chapter 6

UI Principles

Modern.

Minimal.

Responsive.

Accessible.

Fast.

Professional.

No unnecessary animations.

Dark mode supported.

Consistent spacing and typography.

Chapter 7

Backend Principles

Controller

↓

Service

↓

Repository

↓

Database

Controllers never contain business logic.

Services contain business logic.

Repositories access data.

Chapter 8

Database Principles

Normalized where appropriate.

Indexes only when justified.

Soft delete where useful.

Audit columns on important tables.

Proper foreign keys.

Chapter 9

API Principles

RESTful.

Consistent.

Versioned.

Predictable.

Standard response format.

Clear error messages.

Chapter 10

Git Principles

Main

Always stable.

Develop

Integration branch.

Feature branches

One feature per branch.

Small commits.

Meaningful commit messages.

Chapter 11

Documentation Principles

Every module has documentation.

Every API documented.

Every database change documented.

Every architectural decision recorded.

Chapter 12

Testing Principles

Unit Tests.

Integration Tests.

API Tests.

Manual Testing.

No feature is complete until tested.

Chapter 13

AI Provider Rule

Never hardcode an AI provider.

Instead

AIProvider

Today

↓

Ollama

Tomorrow

↓

Gemini

↓

OpenAI

↓

Claude

↓

DeepSeek

No other code changes.

Chapter 14

Storage Rule

Never directly upload to Cloudinary.

Instead

Storage Provider

↓

Local

↓

Cloudinary

↓

AWS

↓

Azure

Chapter 15

Logging Rule

Every important action is logged.

Authentication

AI

Database

Errors

Security

Performance

Chapter 16

Feature Completion Rule

A feature is not complete until:

- ✓ UI
 - ✓ Backend
 - ✓ Database
 - ✓ API
 - ✓ Validation
 - ✓ Security
 - ✓ Tests
 - ✓ Documentation
 - ✓ Code Review
 - ✓ Git Commit
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Chapter 17

Coding Standards

Java

Google Java Style (with project-specific adjustments).

TypeScript

ESLint

Prettier

Consistent naming.

No magic numbers.

Meaningful variable names.

Chapter 18

Performance Goals

Page load

< 2 seconds (typical broadband, cached assets).

API response

Aim for under 500 ms for standard CRUD operations under normal load.

Lazy loading.

Pagination.

Caching where appropriate.

Chapter 19

Accessibility

Keyboard navigation.

ARIA labels where needed.

Color contrast.

Responsive layouts.

Semantic HTML.

Chapter 20

Future Vision

TalentAI should be designed so future developers can add:

- Mobile app
- AI agents
- Billing
- Enterprise SSO
- Multi-language support

without rewriting the entire application.

This Document Is Frozen

Once approved,

the Engineering Constitution should almost never change.

Everything else follows it.
